

PalmSource®

Conference and Expo



True Stories of Palm OS® Enterprise Application Developers

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Learn valuable insights into the issues facing real-world Palm OS® Enterprise Developers



Agenda

- The Real Developers
- The Questions
- The Good & the Not-So-Good
- The Road Ahead



Real Developers

- Responses from developers who are:
  - Working on a real enterprise product
  - Knowledgeable of Palm OS development
  - Employed by key ISV partners
- Responses will be “anonymous” to protect the innocent



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Question #1

" From the point of view of a developer creating enterprise-class applications, what is the best part about Palm OS development? "



Responses to Question #1

- ✓User base – Largest user base for handheld computers
- ✓Clean & Simple User Interface
- ✓Excellent binary emulator w/ Gremlins
- Efficient programs, small apps low overhead
- One chipset to deal with
- Useful documentation



Question #2

“ From the point of view of a developer creating enterprise-class applications, what is the worst part about Palm OS development?”



Responses to Question #2

- Development tools for application developers, too bare-bones & hostile
- Project start-up time, lots of programming overhead
- Maintainability is very poor, apps over 300K need memory protection
- Portability, little code can be ported
- Lack of basic operating system support
- Would like a JAVA VM



Responses to Question #2 cont….

- Inconsistencies between devices
- Want C++ libraries
- Memory segmentation
- Shared Library restriction of 64K
- Command line compiler for CodeWarrior
- Lack of helper classes to aid in UI programming



Question #3

"Which specific Palm OS® development resource did you find the most useful overall, when developing your application (e.g. particular reference book, online knowledge base, Palm developer support, web site, etc)."



Responses to Question # 3

- ✓PDF files from the SDK
- Sample code from the SDK
- O'Reilly programming books
- gcc tutorial and header files
- Knowledge base
- Online Mailing List
- www.palmos.com/dev



Question #4

"What tools (IDEs, compilers, debuggers, languages) did you use for your development, why did you choose these tools, and would you make the same choice again next time?"



Responses to Question #4

- Metrowerks CodeWarrior for Palm OS v8, for debugging features, ideal for component programming.
- PRCTools
- CW not suited for application programming
- Java or AppForge for form based programs
- PilRC (www.ardiri.com)



Question #5

"From the point of view of a developer creating enterprise-class applications, what is the biggest difference between Palm OS development and other types of development you do (desktop, server)?"



Responses to Questions #5

- Memory management is very fragile, slightest mistake would cause crash.
- Fine for component development, a problem for application programmers.
- Limited resources: memory, screen size.
- Slow processor, must optimize for speed.
- Missing threading, concurrent execution, network access.



Question #6

"Looking back, what one thing do you wish existed that could have made your Palm OS development so much easier?"



Responses to Question #6

- C++ application framework
- Faster Technical Support
- Icon and code libraries
- A rich C++ application library
- A server-based synchronization framework
- Easing of code segment and dynamic memory restrictions.



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Summary

The Good News

- Large body of developer resources to help you build Palm OS® applications
- Palm Developer quarterly CD
- Growing list of development tools for component and application developers



Summary

The Not-So-Good News

- Faster processor, improved memory management, multi-threaded OS
- Need for a coherent Java strategy
- More support for C++ development
- Need for better application development tools



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The Road Ahead

- Enterprise development will change as follows:
  - Advanced OS – ARM based
  - Better tools – plug ins to existing IDEs like Visual Studio
  - Java
  - Standard Internet development – Q Browser, plug ins, XML
  - Pushing data to wireless devices



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